IA2 Report

Deep Learning (216U42C601)-IA2

B.Tech AI&DS - TY

Division : A

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Abstract

Solar panels are highly sensitive to environmental factors such as dust accumulation, snow coverage, bird droppings, and physical or electrical damage. These factors significantly reduce the efficiency of solar modules, leading to decreased energy output and increased maintenance costs. Manual inspection of solar panels is time-consuming and inefficient, especially in large-scale solar farms.

This project proposes an automated deep learning-based system for classifying solar panel surface conditions using image data. A multi-class classification model is developed using PyTorch, leveraging transfer learning with EfficientNet-B3. The dataset consists of six classes: Clean, Dusty, Bird-drop, Electrical Damage, Physical Damage, and Snow Covered.

To address challenges such as class imbalance and limited dataset size, advanced techniques including data augmentation, weighted sampling, mixup augmentation, label smoothing, and differential learning rates are employed. The model is further enhanced using a warmup cosine learning rate scheduler and Test-Time Augmentation (TTA).

The proposed approach achieves a validation accuracy of 86.9% and improves test accuracy from 78.7% to 87.23% using TTA. The results demonstrate the effectiveness of deep learning techniques in real-world solar panel monitoring applications.

Introduction

With the increasing adoption of renewable energy, solar panels have become a crucial component of modern power generation systems. However, their performance is significantly affected by environmental contaminants such as dust, bird droppings, snow, and various forms of damage. These factors reduce the efficiency of solar panels by blocking sunlight or degrading panel components.

Traditional inspection methods rely on manual monitoring, which is labor-intensive, time-consuming, and prone to human error. Automated inspection using computer vision and deep learning offers a scalable and efficient alternative.

This project focuses on developing a robust image classification model to automatically detect and classify different types of defects on solar panels. The task is formulated as a multi-class classification problem with six categories. The primary challenges include:

- Class imbalance in dataset distribution
- Visual similarity between certain classes (e.g., dusty vs clean panels)
- Variability in lighting conditions and image quality
- Limited dataset size (~876 images)

To overcome these challenges, a combination of transfer learning, advanced augmentation techniques, and optimized training strategies is employed.

Literature Study

Deep learning, particularly Convolutional Neural Networks (CNNs), has shown remarkable performance in image classification tasks. Pretrained models such as EfficientNet, ResNet, and VGG have been widely used in industrial applications due to their ability to extract hierarchical features from images.

EfficientNet introduces compound scaling, which balances network depth, width, and resolution, resulting in improved accuracy with fewer parameters. Transfer learning using pretrained weights significantly reduces the need for large datasets.

Recent studies highlight the importance of:

- Data augmentation for improving generalization
- Handling class imbalance using weighted loss functions and sampling
- Regularization techniques such as dropout and label smoothing
- Advanced inference techniques like Test-Time Augmentation (TTA)

These approaches collectively improve model robustness and performance, especially in real-world datasets with noise and imbalance.

Methodology

1. Dataset Used

The dataset is obtained from Kaggle and contains images of solar panels categorized into six classes:

- Clean
- Dusty
- Bird-drop
- Electrical Damage
- Physical Damage
- Snow Covered

The dataset consists of approximately 876 images with noticeable class imbalance, where some classes contain significantly fewer samples than others.

Challenges:

- Imbalance ratio of approximately 2.8× between largest and smallest class
- Variability in image resolution and lighting conditions
- Noisy data due to web scraping

To ensure fair evaluation, a **stratified split** is performed:

- Training: 75%
- Validation: 15%
- Test: 10%

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2. Architecture / Workflow

The overall workflow of the system is as follows:

```
INPUT (300×300×3)
     │
     ▼
┌──────────────────────────────────┐
│    EfficientNet-B3 Backbone      │
│    features[0..6]  ← FROZEN      │
│    features[7..8]  ← FINE-TUNED  │  lr=1e-4
└──────────────────────────────────┘
                │
                │ AdaptiveAvgPool → [B,1536]
                ▼
┌──────────────────────────────────┐
│    Custom Head                   │  lr=1e-3
│    Dropout(0.4)                  │
│    Linear(1536→512)              │
│    BatchNorm1d + ReLU            │
│    Dropout(0.2)                  │
│    Linear(512→6)                 │
└──────────────────────────────────┘
                │
                ▼
          Softmax → 6 classes
```

i.e :

Input Image

↓

Data Preprocessing & Augmentation

↓

EfficientNet-B3 (Pretrained Backbone)

↓

Selective Fine-Tuning (Last 2 Blocks)

↓

Custom Classifier Head (FC Layers)

↓

Softmax Output (6 Classes)

Model Architecture
The model is based on **EfficientNet-B3**, chosen for its balance between accuracy and computational efficiency.

Key Design Choices:
- Pretrained on ImageNet
- Backbone layers initially frozen
- Last 2 blocks unfrozen for fine-tuning
- Custom classifier head:

Dropout → Linear (1536 → 512) → BatchNorm → ReLU → Dropout → Linear → Output (6)

Justification:
- Transfer learning reduces training time and improves performance
- Selective fine-tuning preserves learned low-level features
- Batch normalization stabilizes training
- Dropout reduces overfitting

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3. Data Preprocessing & Augmentation

To improve model generalization, extensive augmentation techniques are applied:

Preprocessing:
- Resize images to 300×300
- Normalize using ImageNet statistics

Augmentation Techniques:
- RandomResizedCrop (scale variation)
- Horizontal & Vertical Flip
- Rotation (±30°)
- ColorJitter (lighting variation)
- Gaussian Blur (camera noise simulation)
- Random Grayscale
- Random Erasing (occlusion simulation)

Imbalance Handling:
- WeightedRandomSampler (oversampling minority classes)
- Class-weighted CrossEntropy Loss
- Mixup augmentation ($\alpha = 0.4$)

These techniques ensure the model learns robust and generalized features.

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4. Experiments

Several experiments were conducted to optimize performance:

- Baseline model vs augmented model
- With vs without Mixup
- Weighted loss vs standard loss
- Different learning rates for backbone and classifier
- Standard inference vs Test-Time Augmentation

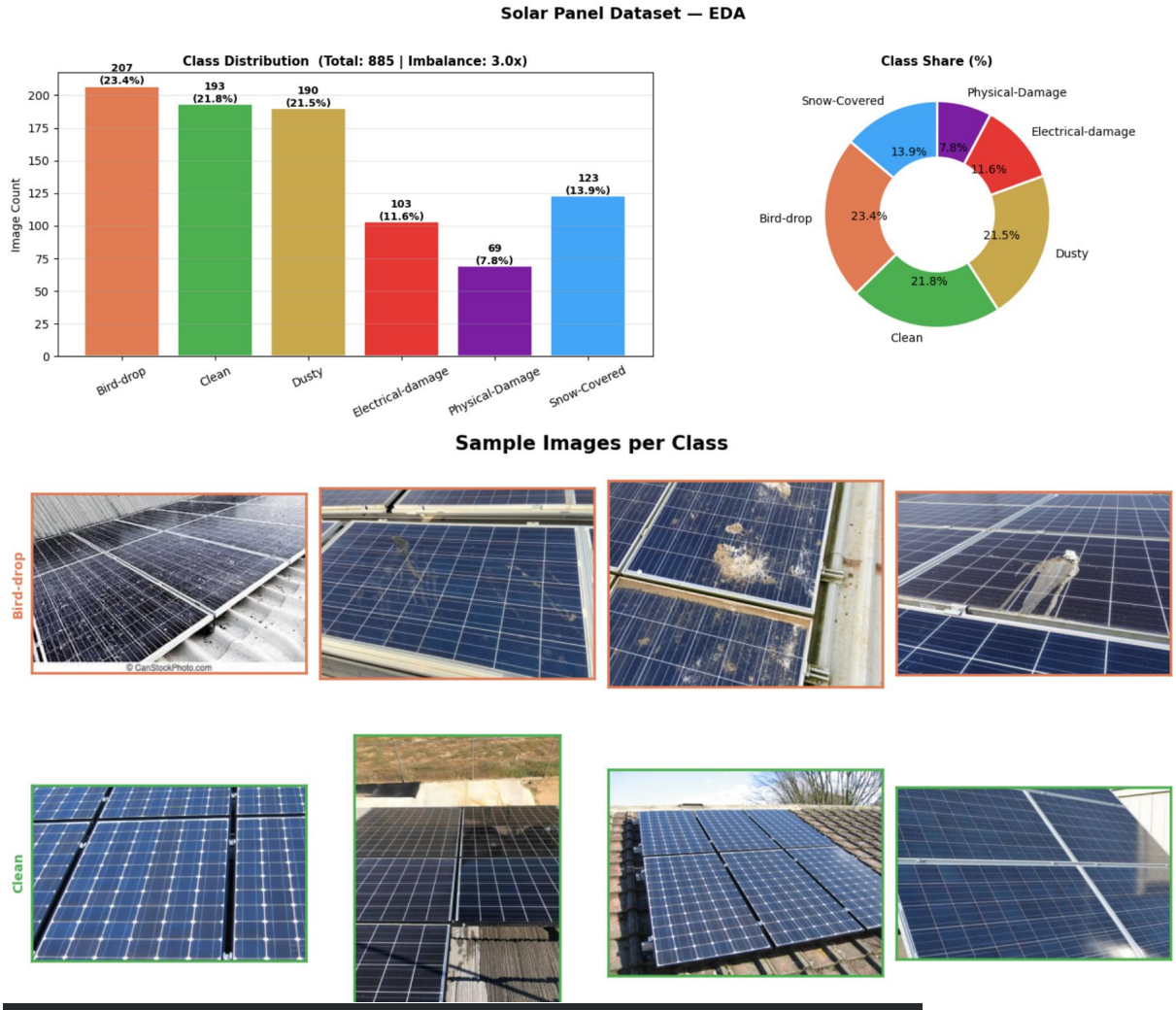
Training Strategy:

- Optimizer: AdamW
- Learning rate:
  - Backbone: 1e-4
  - Classifier: 1e-3
- Scheduler:
  - Warmup (5 epochs)
  - Cosine Annealing
- Regularization:
  - Dropout (0.4)
  - Label smoothing (0.1)
- Early stopping applied

Results and Discussion :

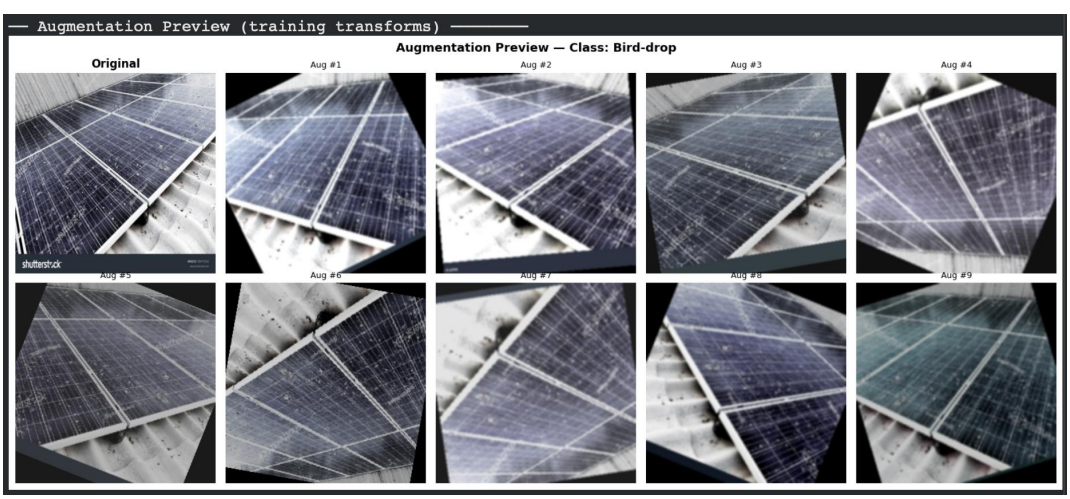
EDA:

```
── Dataset Summary ──────────────────────────────────
  Bird-drop              207
  Clean                  193
  Dusty                  190
  Electrical-damage      103
  Physical-Damage         69
  Snow-Covered           123
  ──────────────────────────────────────────────────
  Total images        : 885
  Imbalance ratio     : 3.00x (Bird-drop vs Physical-Damage)
```

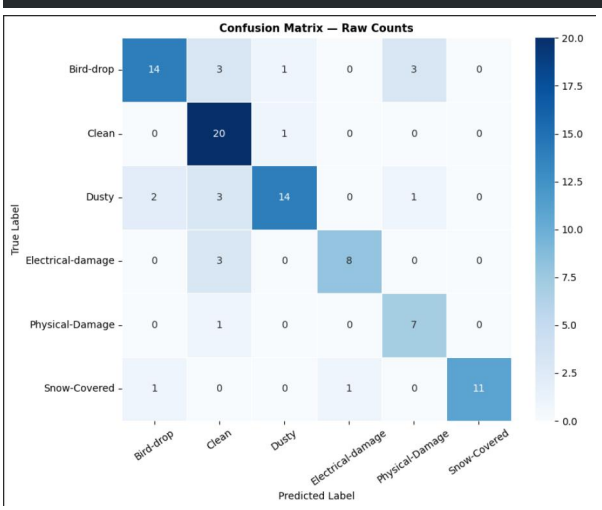


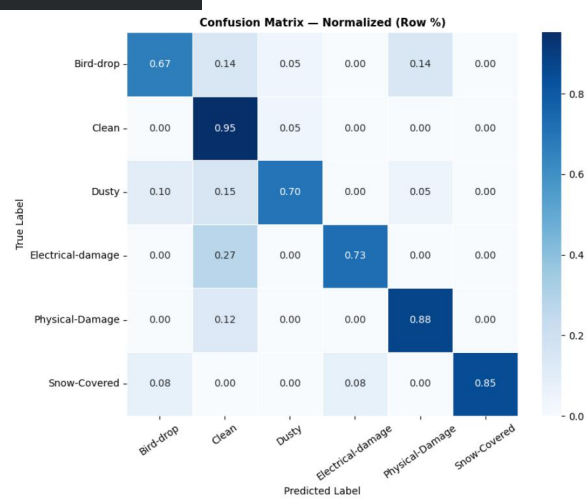

```
── Image Size Statistics (sample of 40/class) ────────
  Width  — min: 165  max:5472  mean:   945  median:   748
  Height — min: 136  max:4032  mean:   764  median:   562

  → We resize everything to 300x300 (EfficientNet-B3 native resolution)
```

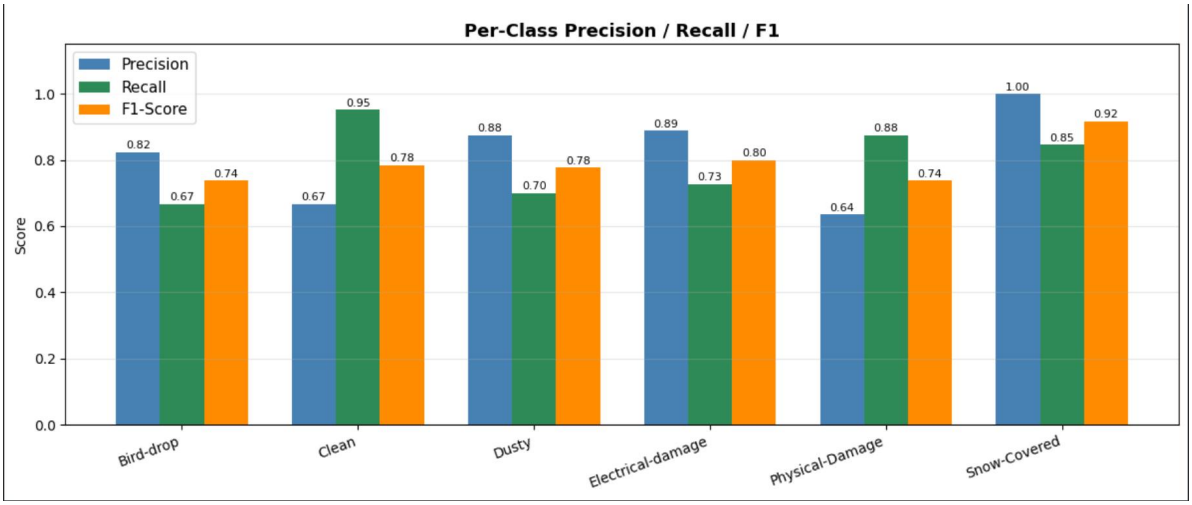
Augmentation:



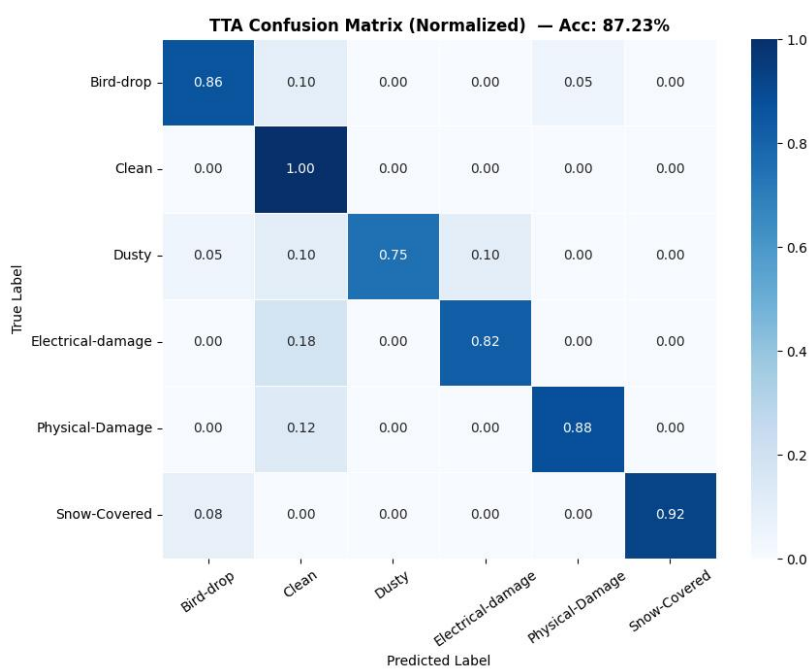
```
Standard Test Accuracy : 78.72%
Standard Test Loss     : 0.9265

Classification Report (Standard Inference):
                   precision    recall  f1-score   support

       Bird-drop     0.8235    0.6667    0.7368        21
           Clean     0.6667    0.9524    0.7843        21
           Dusty     0.8750    0.7000    0.7778        20
Electrical-damage    0.8889    0.7273    0.8000        11
  Physical-Damage    0.6364    0.8750    0.7368         8
     Snow-Covered    1.0000    0.8462    0.9167        13

        accuracy                         0.7872        94
       macro avg     0.8151    0.7946    0.7921        94
    weighted avg     0.8156    0.7872    0.7884        94
```








```
Standard Test Accuracy : 78.72%
TTA Test Accuracy      : 87.23%  (Δ = +8.51%)

Classification Report (TTA):
                   precision    recall  f1-score   support

        Bird-drop     0.9000    0.8571    0.8780        21
            Clean     0.7500    1.0000    0.8571        21
            Dusty     1.0000    0.7500    0.8571        20
Electrical-damage     0.8182    0.8182    0.8182        11
  Physical-Damage     0.8750    0.8750    0.8750         8
     Snow-Covered     1.0000    0.9231    0.9600        13

         accuracy                         0.8723        94
        macro avg     0.8905    0.8706    0.8743        94
     weighted avg     0.8899    0.8723    0.8730        94
```



Performance Metrics :

Metric	Value
Validation Accuracy	86.9%
Test Accuracy	78.7%
TTA Accuracy	87.23%
Macro F1-score	~0.80

Analysis

- The model performs best on **Clean** and **Snow Covered** classes due to clear visual patterns
- Moderate confusion observed between:
  - Dusty vs Clean
  - Bird-drop vs Dusty
- Minority classes such as Physical Damage show slightly lower performance

Impact of Techniques

- **Mixup** improved generalization and reduced overfitting
- **Weighted sampling + weighted loss** improved minority class recall
- **Warmup + Cosine scheduler** stabilized training
- **Test-Time Augmentation (TTA)** improved accuracy by ~8–9%

Confusion Matrix Insights

- High diagonal dominance indicates strong classification ability
- Misclassifications mostly occur between visually similar classes
- Electrical damage occasionally confused with clean due to subtle patterns

Limitations

- Limited dataset size
- Class imbalance still affects performance
- Model performs classification only (no localization)
- Dataset may not fully represent real-world conditions

Conclusion

This project successfully demonstrates the application of deep learning for automated solar panel defect classification. By leveraging EfficientNet-B3 with transfer learning and incorporating advanced techniques such as Mixup, weighted loss, and Test-Time Augmentation, the model achieves strong performance despite dataset limitations.

The results highlight the effectiveness of combining robust preprocessing, optimized training strategies, and advanced inference methods in solving real-world computer vision problems.

Future Scope

- Implement Grad-CAM for model interpretability
- Use object detection models (e.g., YOLO) for defect localization
- Expand dataset for improved generalization
- Deploy model on edge devices for real-time monitoring
- Explore Vision Transformers (ViT) for improved performance

References

1. Kaggle Solar Panel Dataset
2. PyTorch Documentation
3. EfficientNet: Rethinking Model Scaling for CNNs
4. Mixup: Beyond Empirical Risk Minimization
5. Research papers on CNN-based defect detection

Github link:

https://github.com/rishimane-svg/DL_IA2_16014224811_16014224812